INCH-POUND

MIL-DTL-85097/15A (OS)
28 November 2005
SUPERSEDING
MIL-A-85097/15 (AS)
27 May 1987

DETAIL SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTOR, MARK 121 MOD 0

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Mark 121 Mod 0 Rocket Motor described herein shall consist of this specification and the latest issue of MIL-DTL-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Mark 121 Mod 0 Rocket Motor (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents specified in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-85097	USN/USMC Propellant Actuated Devices (PADs), General Specification For

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Products Branch (Code E143D), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/

AMSC N/A FSC 1377

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Naval Air Systems Command (CAGE Code 30003)

AS 3885	Propellant, N-47 (U)
673AS300	Rocket Motor, Mark 121 Mod 0

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center. Attn: Technical Information Products Branch (Code E143), Indian Head, MD 20640-5035.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Mark 121 Mod 0 Rocket Motor shall conform to Drawing 673AS300.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-DTL-85097, the propellant shall meet the requirements of Table I.

TABLE I. Propellant N-47 (AS 3885).

Property	Temp (°F)	Zero time		7-Year Equivalent			
				0% Rel. Humidity		22% Rel. Humidity	
		Min	Max	Min	Max	Min	Max
S_m (maximum stress) psi	77	149	—	149	—	149	—
E_m (elongation at Sm) %	77	21.3	—	10	—	18.5	—
Y_m (Young's modulus) psi	77	960	1800	960	2862	960	2880
B_r (strand burn rate at 1000 psi) in/sec	77	0.69	0.84	0.69	0.89	0.69	0.89

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the requirements of AS 3885 when tested in accordance with MIL-DTL-85097.

3.3.2 Ballistic performance. The ballistic performance shall conform to Table II.

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TABLE II. Ballistic performance requirements.

Characteristic	Value Min	Value Max	Temperature (± 5°F)
Rocket motor impulse (lb-s)	30.9	—	-40, 160
Action time (ms)	—	200	-40, 160
Peak thrust (lb)	158	472	-40, 160
Ignition delay (ms)	—	20	-40, 160
Initiation pressure (psig)	400	600	-40, 160
Burn time (ms)	72	—	-40, 160
Impulse Angle (deg)	77	83	-40, 160

4. VERIFICATION

4.1 Preproduction/production

4.1.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: –40° ± 5°F
b. Hot: 160° ± 5°F

4.1.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: Not less than 4 hours
b. Temperature cycling: Not less than 5 hours

4.1.3 Radiographic inspection. All components or assemblies shall be radiographed in one view.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The rocket motor is intended to insure proper trajectory divergence of the aircrew during an escape for the TAV-8B Aircraft Escape System. The rocket motor was designed specifically for use with the TAV-8B Aircraft Escape System in a military aircraft, and has no commercial applications.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

6.3 Definitions. (See Figure 1 for a graphical representation of some of the definitions provided below.)

6.3.1 Peak thrust. Peak thrust is the maximum value of thrust that occurs during the action time interval.

6.3.2 Initiation pressure. Initiation pressure is the pressure at which the shear pin of the initiation assembly shears. This is represented as the point prior to the break in the pressure-time trace from the input pressure signal used to initiate the device.

6.3.3 Impulse angle. Impulse angle is the angle obtained by vectorially adding the vertical and horizontal impulses.

6.3.4 Action time. Action time is determined from the thrust-time trace, and is defined as the time interval from the 10 percent of maximum thrust on the initial rise of the curve to the corresponding 10 percent of maximum thrust on the declining portion of the curve.

6.3.5 Burn time. Burn time is the time between the initial 10 percent maximum thrust level and the web burnout point as indicated by the sharp downward break on the thrust-time trace.

6.3.6 Rocket motor impulse. Rocket motor impulse is obtained by integrating the resultant thrust over the action time.

6.3.7 Ignition delay. Ignition delay is the elapsed time from the ignition signal (shear pin shears) to 10 percent of maximum thrust, as measured on the rising portion of the rocket motor thrust trace.

6.4 Subject term (keyword) listing.

TAV-8B
N-47

6.5 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. The changes in this revision were authorized by ECP 05037-5110L.

Custodian:
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Preparing activity
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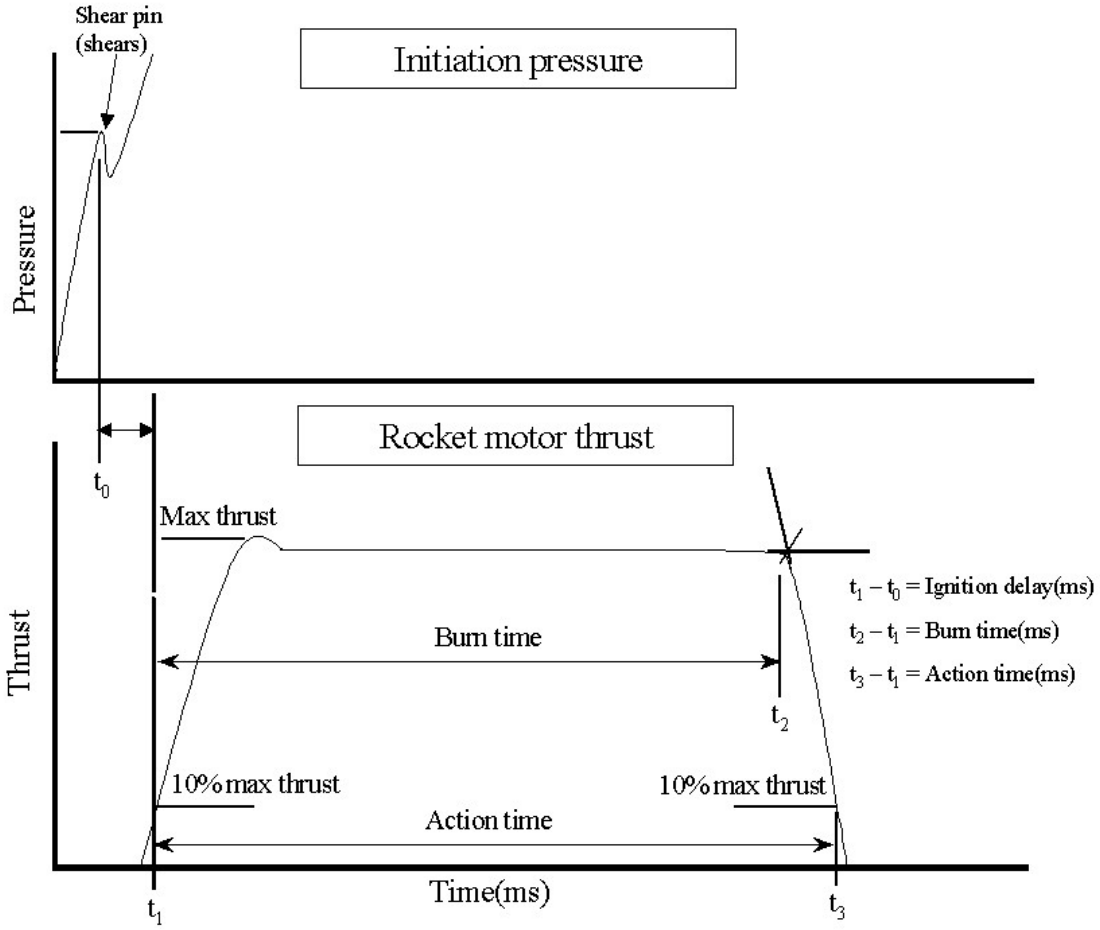


FIGURE 1. Sample performance trace.

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